READING GUIDE

Narrative-Behavior Measurement

Full results | CEO speech, R&D prediction and dividend policy | 31 October 2026

This report brings together the existing empirical results for all four steps. It preserves the supplied scientific specification and distinguishes reported findings, archived evidence and unresolved questions. It is a results publication; the companion replication package documents computational reruns.

The central score is text-predicted R&D divided by assets. It is a candidate behavioral proxy. Predictive fit and regression significance alone do not establish that it measures psychological risk preference.

Step	What is reported
1. Represent	354 topics; original descriptions, training metadata and catalogue
2. Screen	12 trials, 36 readers; both stages, missingness and recoveries
3. Predict	29,506 firm-year scores; Ridge/XGBoost and six robustness runs
4. Assess	25 downstream models, including 16 score specifications

Read the report first. Then open the step folders for full CSV/JSON outputs and original Excel workbooks. Every retained result is indexed with a SHA-256 fingerprint. The complete results ZIP additionally includes the 3,840-file original reader-evidence archive, already expanded for inspection.

Scope: the completed 21 October screening pilot and empirical packages delivered on 27 October 2026. Earlier design proposals and the 19 October pilot are not substituted for this completed application.

STEP 1 / REPRESENT

An existing 354-topic representation

The supplied BERTopic model was trained on 887,488 speech components with nonmissing political-donation ideology scores, drawn from a 5,765,638-component 2013-2023 corpus. That historical training sample is not restricted to CEOs.

The saved model contains 354 substantive topics (IDs 0-353), plus outlier class -1. There are 482,860 outlier training components (54.4%). The current application uses 1,775,355 CEO-tagged components and the supplied topic densities.

Component	Delivered setting or output
Embedding	all-mpnet-base-v2; 768 dimensions
UMAP	15 neighbors; 5 dimensions; cosine; random_state 42
HDBSCAN	min_cluster_size 120; min_samples 100; eom
Topic descriptors	Names, sizes, keywords, representatives; Excel and CSV
Model artifacts	topics.json and configuration retained in this results package

The safetensors save does not preserve the fitted UMAP and HDBSCAN objects. A new training run can produce a different topic system; it is not a way to recover missing historical ordering evidence.

The historical density matrix has no embedded row identities or model/column fingerprint. Its ordering remains delivery-declared. Text joins and subsequent input checks should not be described as independent authentication of that historical matrix.

Source: Topic descriptions workbook - step1/topic_descriptions_354Topics.xlsx

Source: Topic information CSV - step1/topic_info_354Topics.csv

Source: Exact screening catalogue - step1/catalogue.json

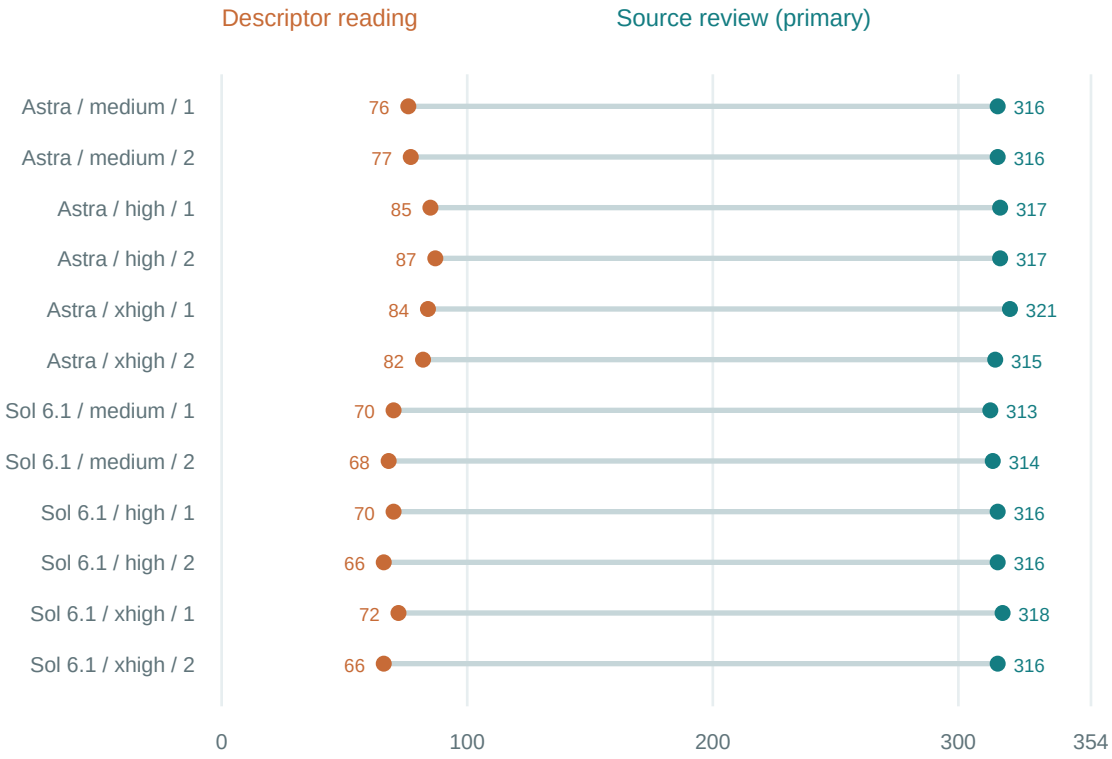
Source: Original training method - step1/methods/1_topic_model_training.md

STEP 2 / SCREEN

Source review changes the selection

Twelve P0 trials cross two requested model families, three effort settings and two repetitions. Each trial has three readers. Stage A uses descriptors; Stage B continues each reader through source review. An unweighted two-of-three majority determines selection.

Stage A selected 66-87 topics per trial. Stage B selected 313-321. The figure shows the primary first-attempt results; the growth is sensitivity to the reading procedure, not an accuracy estimate.



Source: All counts and coverage - step2/analyses/primary/trial_selection_counts.csv

Source: Topic-by-topic reader browser - step2/explorers/index.html

STEP 2 / EVIDENCE

Keep missingness and minority reasons visible

The primary analysis retains first saves. Six malformed or failed page saves across five readers leave 120 Stage B rows absent. There are 12,624 observed Stage B topic rows out of 12,744. Supplementary recovery records fill the roster without replacing the primary analysis.

Across 4,248 trial-topic decisions, primary totals are 3,795 relevant, 452 irrelevant and one unresolved. Supplementary totals are 3,796, 452 and zero unresolved. The unresolved primary case is topic 224 in repetition 2, Astra Xhigh.

Among 4,247 jointly resolved Stage A-to-B majority comparisons, 2,964 change (69.79%): 2,928 irrelevant-to-relevant and 36 relevant-to-irrelevant. There are 306 topics always selected, 31 always excluded and 17 that vary in the primary Stage B outputs.

Evidence or qualification	Retained result
Primary complete Stage B rosters	31 of 36 readers
Primary mechanical flags	62 (coverage, quotation, keyword and roster flags)
Supplementary mechanical flags	12; diagnostics are not semantic error counts
Recoveries	51 recovery pairs; 52 supplementary page attempts
Scope	12 P0 trials ratified after collection; no winning classifier established

The delivered Step 3 input uses primary b1-gpt-6-astra-P0-xhigh: 321 selected, 33 excluded and none unresolved. This identifies the analysis actually supplied; it does not establish superiority over the other trials.

The browser exposes all reader reasons, confidence explanations, quotations and missing rows. The full archive also preserves literal attempts, seals, provenance, recoveries and the documented protocol deviation.

Source: Primary analysis report - step2/analyses/primary/REPORT.md

Source: Supplementary analysis report - step2/analyses/supplementary/REPORT.md

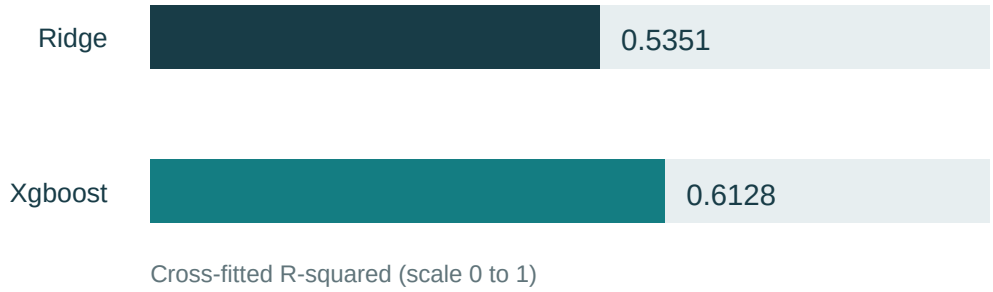
Source: All reader record locations - step2/READER_INDEX.json

STEP 3 / PREDICT

Text predicts R&D intensity

The model uses 321 selected topic features and one aggregate sentiment feature. Topic features receive a centered log-ratio transformation and standardization. Firm-CEO connected components define grouped five-fold cross-fitting.

The panel contains 29,506 firm-years, 4,517 firms and 6,933 CEOs over 2013-2023. Targets are available for 29,191 rows. Missing R&D is set to zero when assets are positive; 42.7% of available targets arise from this zero-fill rule. The other 315 rows lack a target because assets are missing or nonpositive.



Delivered metric	Ridge	XGBoost
Cross-fitted R-squared	0.5351	0.6128
Mean squared error	0.006006	0.005001
Correlation with target	0.7315	0.7829

The reported tuning choices are Ridge alpha = 1,000 and XGBoost max_depth = 6. Evaluation uses the delivered target-available rows and winsorization convention. Per-fold counts, settings and bounds are retained in metrics.json.

The score files contain out-of-fold and full-sample fitted versions plus clipped, percentile and standardized transformations. Downstream analysis uses out-of-fold scores. No industry, size or year controls enter this prediction model. This specification differs from the earlier proposed scalar N followed by f(N, controls).

Source: All 29,506 firm-year scores (CSV) - step3/results/risk_scores.csv

Source: Scores (Parquet) - step3/results/risk_scores.parquet

Source: Complete model and fold metrics - step3/results/metrics.json

Source: Full method - step3/docs/METHOD.md

STEP 3 / ROBUSTNESS

The supplied small-training-sample checks

The delivered robustness exercise uses three random seeds. Each seed forms a grouped 20% training / 20% validation / 60% test split. Hyperparameters are selected on the validation set; test results are compared with the supplied out-of-fold scores on the same rows and target convention.

These six runs describe sensitivity to the supplied split design and training size. They are not six independent validations of psychological meaning.

Seed	Model	Training N	Test N	Test R-sq	OOF comparison R-sq
1	ridge	5833	17499	0.5057	0.5315
1	xgboost	5833	17499	0.5788	0.6086
2	ridge	5890	17600	0.4960	0.5209
2	xgboost	5890	17600	0.5661	0.5959
3	ridge	5819	17444	0.5183	0.5365
3	xgboost	5819	17444	0.5824	0.6106

The table reproduces the saved metrics without retraining. Training and test counts, selected hyperparameters, correlations, MSE, training fit and validation fit remain in the complete CSV. The OOF comparison uses models trained on approximately 80% of the available data and is not an equal-training-size benchmark.

Source: All robustness metrics - step3/results/seed_robustness_metrics.csv

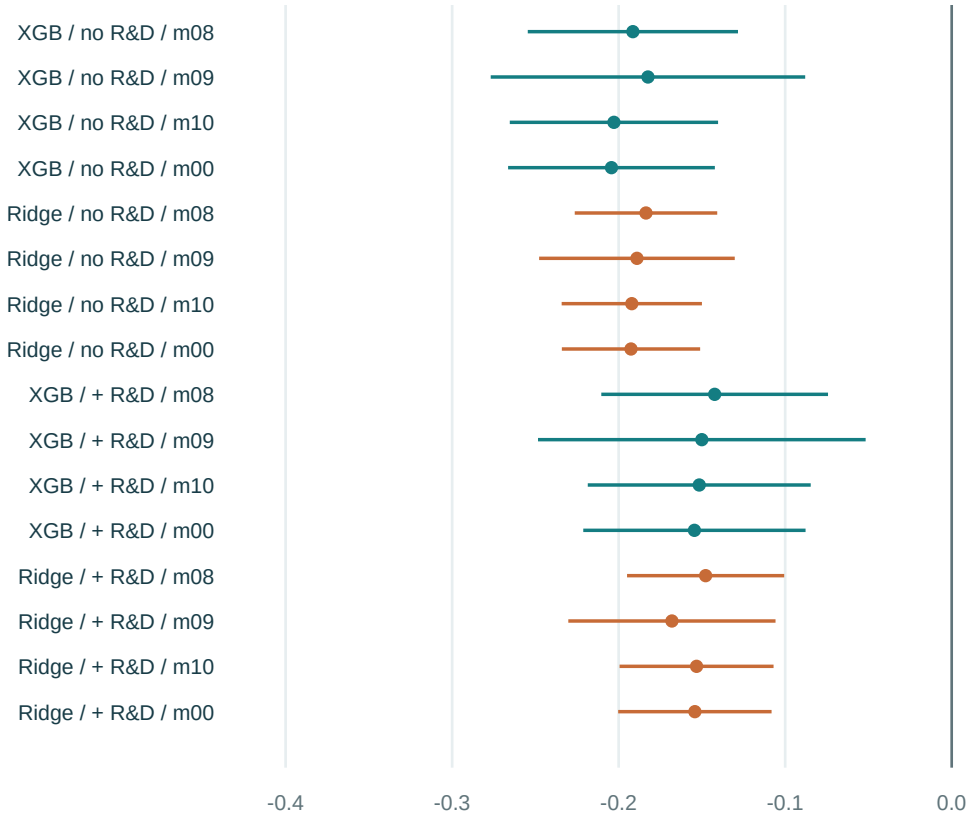
Source: Original robustness report - step3/results/seed_robustness_report.md

STEP 4 / ASSESS

Dividend associations are consistently negative

Strict company, person and timing matching yields 2,967 input observations across 600 firms, from an eligible pool of 7,086 firm-years. Ordinary score models retain 2,927 observations across 591 firms after removing industry groups that perfectly predict dividend status. The positive-relative-leverage specification uses 1,653 observations across 340 firms.

Pooled Logit models include separate industry and year effects and company-clustered standard errors. Score = 100 x max(out-of-fold predicted R&D/assets, 0). Four specifications per algorithm, with and without actual R&D controls, give 16 score models. All supplied score coefficients are negative with clustered p < .01.



Dots show coefficients; lines show 95% Wald intervals (coefficient +/- 1.96 x company-clustered SE). Labels m00, m08, m09 and m10 refer to the supplied specifications. XGB denotes XGBoost.

For XGBoost m08 with actual R&D controls, the coefficient is about -0.1423 (SE 0.0347), implying an odds ratio near 0.867 per one-percentage-point increase in the score. This is a conditional log-odds association, not a probability-point change or causal effect.

Source: Full score-model results - step4/score-models/results/RESULTS.html

Source: All regression coefficients - step4/full/estimates/coefficients.csv

STEP 4 / COMPLETE INVENTORY

All 25 supplied model specifications

This inventory includes the nine baseline/replication specifications and all 16 score specifications. The original five-sheet workbook, full tables, estimates and used-sample records accompany the report. The table gives score coefficients only; full control and fixed-effect coefficients are retained in the coefficient CSV.

Model ID	N used	Firms	Score coef	Clustered SE
rep_original_2006_2011_m08	5736	1424	See full table	-
rep_original_2006_2011_m09	3416	850	See full table	-
rep_original_2006_2011_m10	5736	1424	See full table	-
rep_extended_2006_2023_m08	13716	1793	See full table	-
rep_extended_2006_2023_m09	7784	1055	See full table	-
rep_extended_2006_2023_m10	13716	1793	See full table	-
rep_2013_2023_m08	7035	1304	See full table	-
rep_2013_2023_m09	3782	728	See full table	-
rep_2013_2023_m10	7035	1304	See full table	-
score_xgboost_no_rd_m08	2927	591	-0.1915	0.0322
score_xgboost_no_rd_m09	1653	340	-0.1824	0.0482
score_xgboost_no_rd_m10	2927	591	-0.2028	0.0319
score_xgboost_no_rd_m00	2927	591	-0.2043	0.0317
score_ridge_no_rd_m08	2927	591	-0.1836	0.0218
score_ridge_no_rd_m09	1653	340	-0.1890	0.0300
score_ridge_no_rd_m10	2927	591	-0.1921	0.0215
score_ridge_no_rd_m00	2927	591	-0.1926	0.0212
score_xgboost_with_rd_m08	2927	591	-0.1423	0.0347
score_xgboost_with_rd_m09	1653	340	-0.1501	0.0502
score_xgboost_with_rd_m10	2927	591	-0.1516	0.0342
score_xgboost_with_rd_m00	2927	591	-0.1545	0.0341
score_ridge_with_rd_m08	2927	591	-0.1478	0.0241
score_ridge_with_rd_m09	1653	340	-0.1680	0.0317
score_ridge_with_rd_m10	2927	591	-0.1532	0.0236
score_ridge_with_rd_m00	2927	591	-0.1542	0.0235

Source: Original complete workbook - step4/full/results/table4_annual_b_tables.xlsx

Source: All 25 model tables - step4/full/results/REPLAYED_TABLES.html

Source: Model summary and diagnostics - step4/full/estimates/model_summary.csv

Source: Model specifications - step4/full/estimates/model_reference.json

INTERPRETATION & FILE GUIDE

What the results establish, and what remains open

The supplied application connects a recorded topic-selection procedure, grouped prediction of a behavioral target, and a completed dividend analysis. The evidence supports those computational and associational descriptions.

Psychological meaning remains a research question. The score may reflect industry, opportunity sets, firm resources and disclosure practices as well as risk preference. Human-reference screening accuracy and a separate N/B/S construct-validation comparison are not established.

Interpretation also depends on historical density alignment, zero-filled R&D, overlap between speech and target timing, topic fitting that included held-out firms' text, and selective downstream matching. Clustered regression standard errors condition on frozen scores; they do not propagate first-stage retraining uncertainty. Related specifications are not independent replications.

Inspect	Open within the results folder
Report and figures	report.html / report.pdf / figures/
All topics and descriptors	step1/
All 12 trials and all reasons	step2/explorers/index.html
Firm-year scores and robustness	step3/results/
All downstream tables and estimates	step4/full/
Original literal reader evidence	reader-evidence/ (complete ZIP only)
File identities and source mapping	MANIFEST.json / SOURCES.json

This report repackages existing findings; it does not claim a new empirical rerun. Newly executed reproduction evidence belongs to the separate replication package. Historical audit files remain explicitly identified as supplied checks.

All files are shared within the private research repository under the source-data access conditions. Original numerical outputs and reader judgments are retained unchanged; figures, navigation and reader browsers are new presentations of those saved outputs.